USER MANUAL

GMC Radiation Monitor

Home Assistant Add-on - Version 8.4.2

For beginners, users and administrators

July 2026 edition

About this manual

This manual explains the GMC Radiation Monitor add-on in plain language. It covers installation, daily use, analysis, history, reports, backups and examples for individual sensors.

!	**Sample data** All screenshots and values are demonstration data. Device names, entity IDs, serial numbers, calibration values and GMCMap IDs must be adapted to your own installation.
!	**Important safety information** The add-on is intended for monitoring and home automation. It is not a calibrated radiation-protection instrument and does not replace official measurements, professional advice or instructions from authorities.

How to use this manual

- New installation: read chapters 1 to 4 first.
- Daily use: chapters 4 to 9 explain the dashboard.
- Unusual values or errors: use chapters 2 and 12.
- Administration: chapters 8 to 11 cover automation, exports, backups and integrations.

Contents

Home Assistant add-on - Version 8.3.2

1. Quick start

Use these steps to reach a clear first dashboard without changing advanced settings.

1. Connect the compatible GMC counter to the Home Assistant host by USB.
2. Install and start the add-on. Open the web interface.
3. Confirm that the device is online and that the latest CPM value updates.
4. Select Summary first. Change to Analysis only when you need more detail.
5. Let the system collect measurements before judging trends or the local baseline.

!	**Tip** A new installation needs time to learn the normal local background. Short-term fluctuations are normal.

2. Safety and measurement basics

CPM means counts per minute. The value depends on the detector tube, device model and environment. Compare a device mainly with its own history and configured safety limits.

- Absolute safety limits and the local background analysis answer different questions.
- A green local-background card does not override an absolute warning.
- A single spike should be checked, but a sustained rise is more important.
- Keep distance from an unknown source and follow local emergency guidance when a real hazard is suspected.

!	**Important safety information** Never copy calibration factors or warning limits from an example without checking the device documentation and your intended use.

3. Installation and first start

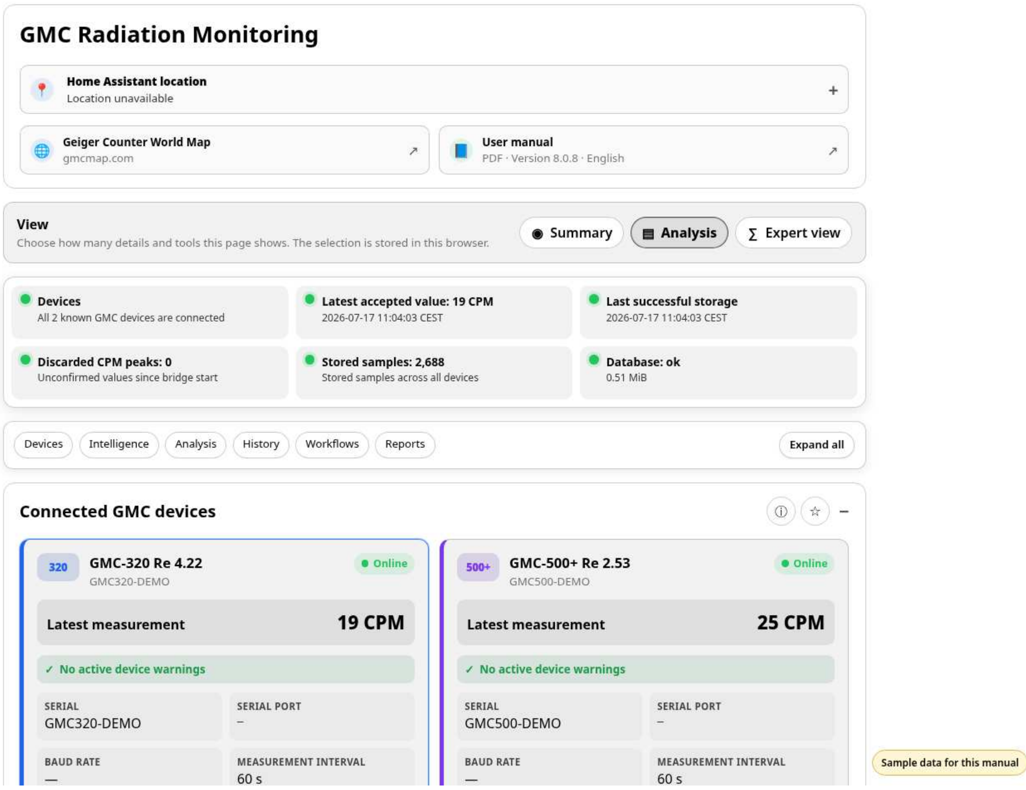
The add-on automatically detects supported serial devices. Stable device paths are preferred so that reconnection works after a restart.

1. Install the add-on package and review the example configuration.
2. Adapt device names, sensor entity IDs and optional GMCMap settings.
3. Start the add-on and check the log for device discovery.
4. Open the dashboard and verify the status overview.
5. Create a managed backup before larger configuration changes.

!	**Info** If no device appears, check the USB cable, power supply, host permissions and whether another process is using the serial port.

4. Dashboard overview

The upper area gives the fastest overview. It contains the location, resource links, view selector, status summary and navigation.

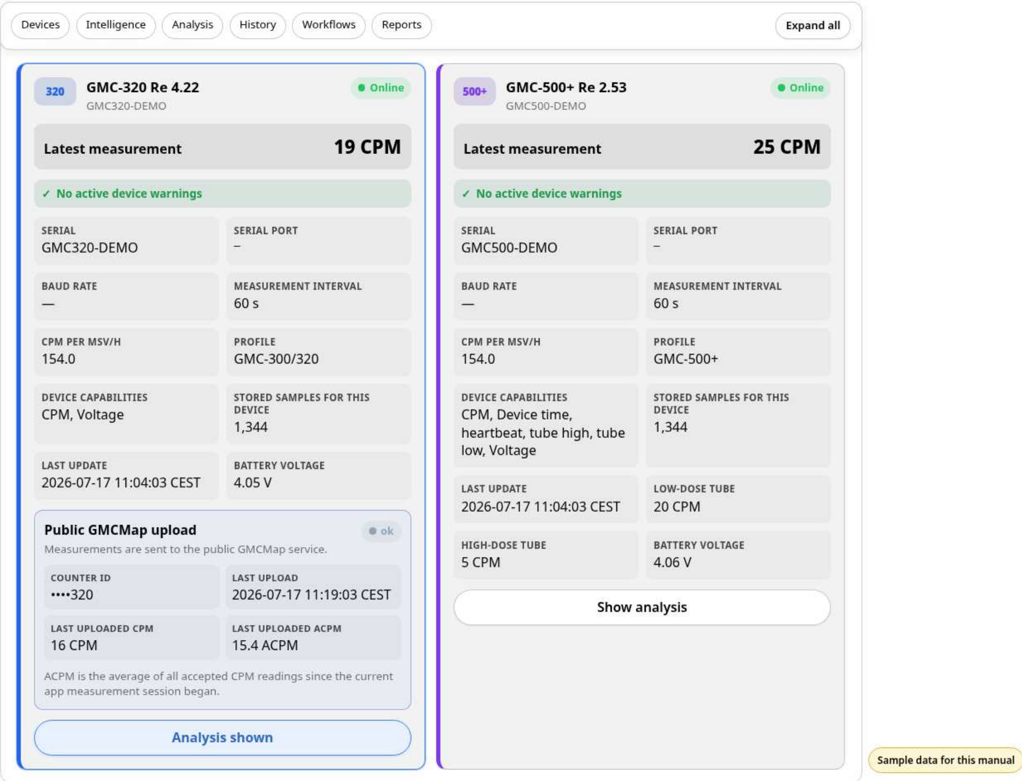


4. Dashboard overview

- The World Map link opens GMCMap. The manual link opens the language that matches the dashboard.
- Summary, Analysis and Expert view control how much detail and how many tools are shown.
- The status summary shows connectivity, the latest accepted value, storage status and database health.
- Navigation jumps directly to devices, analysis, history, workflows and reports.

5. Views and devices

Every device has its own card. The card shows availability, the latest value, warnings, identity, serial connection and device-specific options.



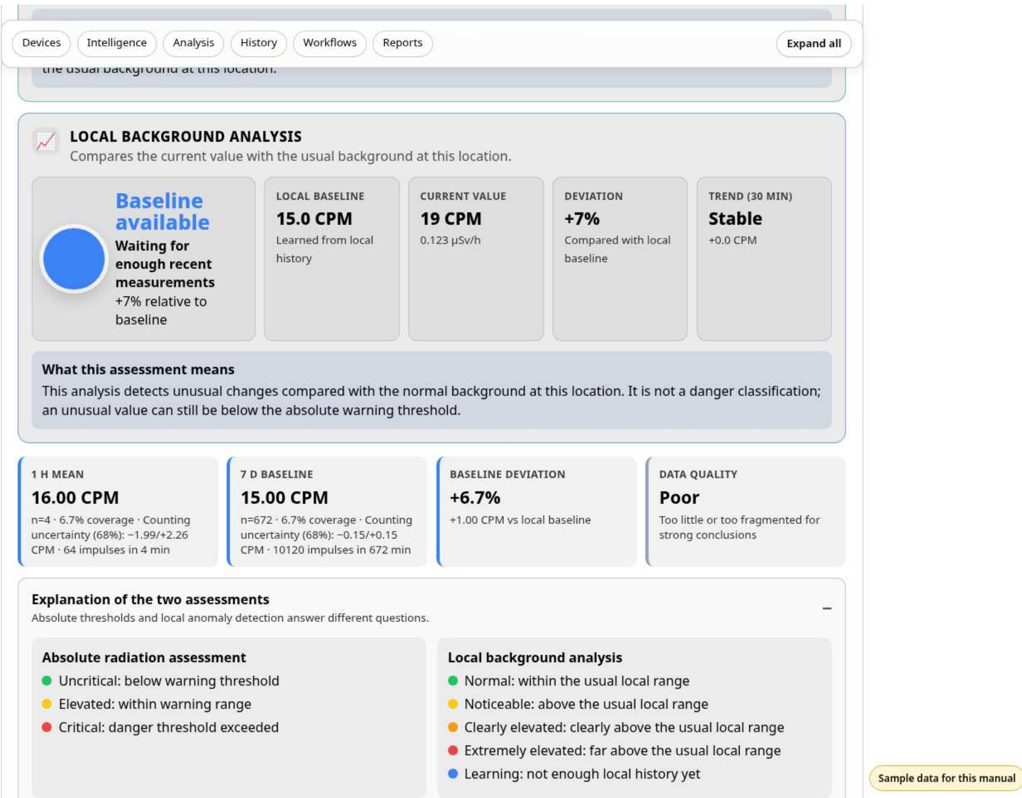
5. Views and devices

- Use clear device names that describe the room or purpose.
- Check the online state before interpreting a missing or old measurement.
- Device warnings are more important than decorative status colors.
- Expand technical details only when diagnosing a connection or configuration issue.

!	**Tip** The selected view is stored in the browser. Different computers can therefore use different detail levels.

6. Intelligent evaluation and analysis

The analysis combines absolute limits, local background, trends, data quality and optional environmental information. It is designed to explain why a value is considered normal or unusual.



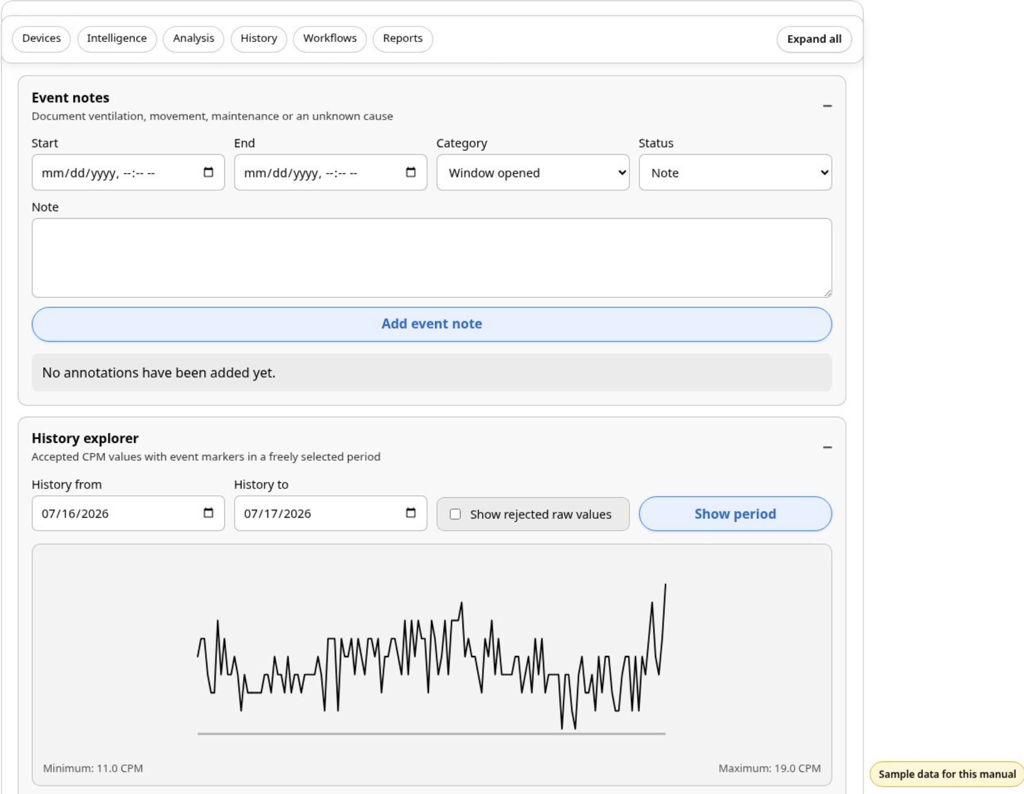
6. Intelligent evaluation and analysis

- Summary gives the recommended everyday view.
- Analysis adds reasoning, trend cards and background comparisons.
- Expert view reveals raw diagnostics, quality information and additional exports.
- Read the recommendation together with the underlying cards; do not rely on one number alone.

!	**Info** “Unremarkable” means that the value fits the learned local pattern. It is not a general declaration that radiation is harmless.

7. History and event notes

History shows changes over time. Event notes help explain ventilation, maintenance, moved devices or other circumstances that can affect the data.



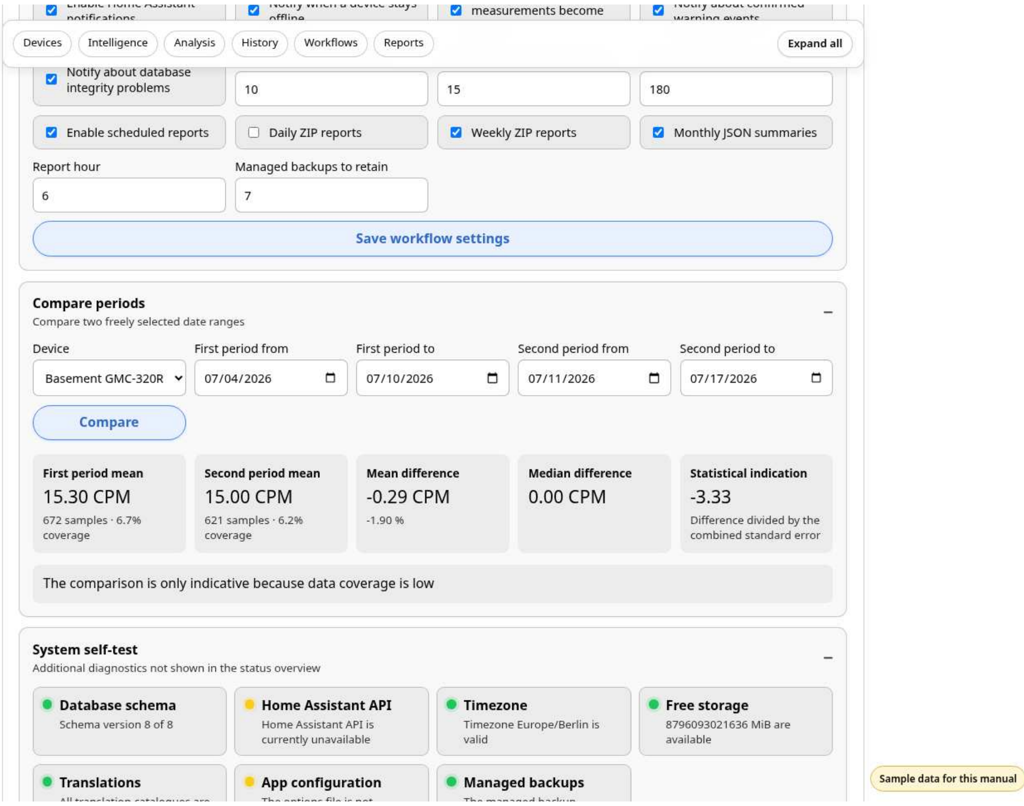
7. History and event notes

1. Choose the device and time range.
2. Compare the graph with the learned baseline and event notes.
3. Add a note when you move a device, change a cable or alter the room conditions.
4. Use period comparison to distinguish a lasting change from a short event.

!	**Tip** Good event notes make later analysis much easier. Use short factual descriptions and exact times.

8. Workflows and notifications

Workflows automate routine tasks without changing the measurement process. Available options include notifications, scheduled reports and period comparisons.



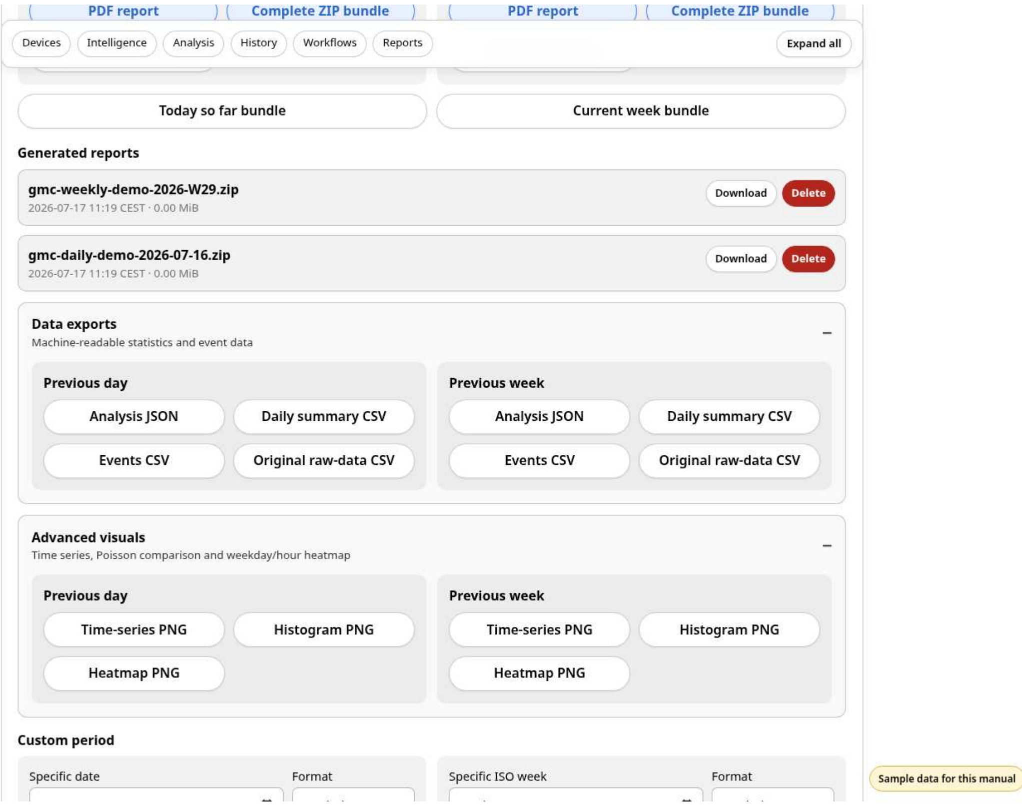
8. Workflows and notifications

- Enable only notifications that someone will actually receive and act on.
- Test the offline threshold by disconnecting a non-critical test device.
- Choose a scheduled report time when Home Assistant is normally running.
- Collapsed sections remember their state in the browser.

!	**Important safety information** Automation is an aid. It does not guarantee delivery and must not be the only safety measure.

9. Reports, exports and backups

Reports provide understandable summaries and technical exports. Managed backups protect the application database and settings used by the add-on.



9. Reports, exports and backups

- Use “Measurements as CSV” for quality-filtered values.
- Use “Original raw data as CSV” only when you need expert-level diagnostics.
- ZIP reports bundle multiple files for archiving or sharing.
- Delete generated reports only after confirming that no copy is still required.
- Create a managed backup before upgrades or major option changes.

!	**Info** A backup is useful only when it can be found and restored. Document the storage location and retention policy.

10. Configuration examples

The included configuration is intentionally an example for individual sensors. Replace the example entities and device-specific values with those from your own Home Assistant installation.

```
devices:
  - name: "Basement counter"
    serial_number: "YOUR-SERIAL"
    external_temperature_entity: "sensor.basement_temperature"

system:
  ui_language: "auto"
  timezone: "Europe/Berlin"
```

- Use valid Home Assistant entity IDs.
- Keep serial numbers unique.
- Treat calibration values as device-specific data, not universal defaults.
- Restart the add-on after configuration changes that affect device discovery.

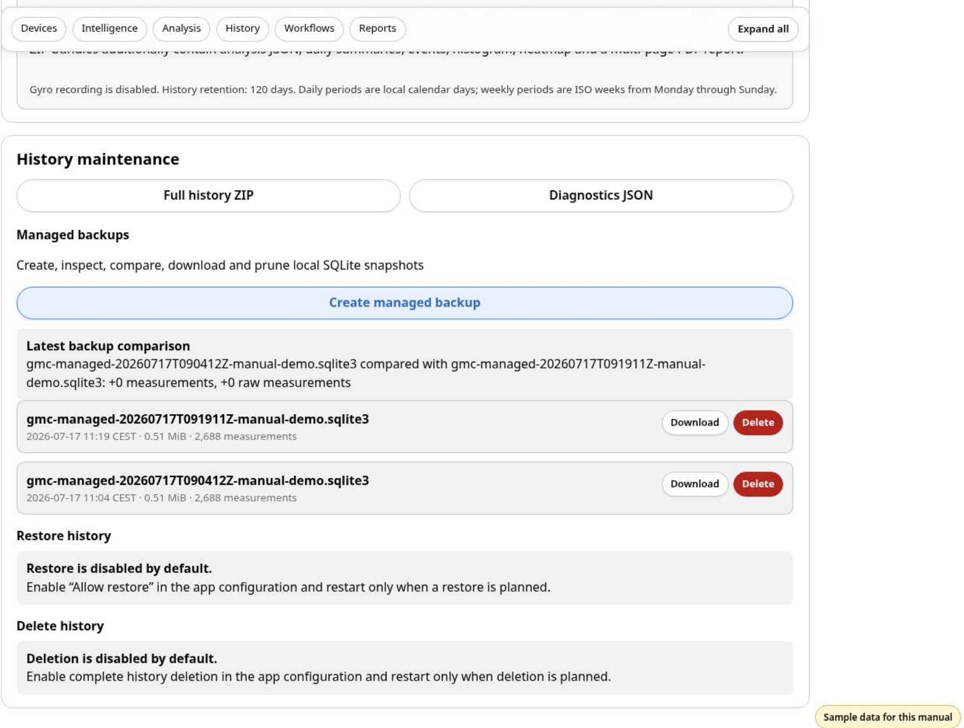
!	**Sample data** All screenshots and values are demonstration data. Device names, entity IDs, serial numbers, calibration values and GMCMap IDs must be adapted to your own installation.

11. GMCMap, Home Assistant and MQTT

GMCMap publishing, MQTT entities and Home Assistant sensors are optional integrations. Configure them only when you understand where the data is sent and how it is used.

- GMCMap can publish selected measurements to the public world map. Check privacy and location settings first.
- MQTT entities expose current values, availability and analysis results to Home Assistant.
- Use Home Assistant automations for convenience, but keep the add-on's own status and reports available for diagnosis.
- Protect API keys and counter identifiers; do not place secrets in screenshots or public issue reports.

!	**Tip** Start with local monitoring. Add external publishing only after the local installation is stable.



11. GMCMap, Home Assistant and MQTT

12. Troubleshooting, glossary and checklist

Most problems can be narrowed down by checking the status overview, device card, log and self-test in that order.

Problem	Action
No device	Check USB, power, permissions and serial-port conflicts.
Old value	Check online state, scan interval and recent log messages.
Unexpected spike	Compare with history, event notes and data-quality information.
No report	Check workflow settings, time zone, storage and free space.
Manual missing	Reinstall the complete 8.3.2 package and refresh the dashboard.

Glossary

Term	Meaning
CPM	Counts per minute reported by the detector.
Baseline	Learned normal range for this device and location.
Trend	Direction of recent measurements over a selected period.
Raw data	Unfiltered records kept for technical analysis.
Managed backup	Application-created backup with retention management.

Final check

- ☐ Device online and value current
- ☐ No active device warning
- ☐ Database status OK
- ☐ Time zone correct
- ☐ Backup current
- ☐ Notification path tested

Version 8.3.4 addendum

This version extends the device view and metrology configuration for detectors with one or two Geiger-Mueller tubes.

Detector and calibration on the main dashboard

- The device card now shows the tube model, conversion factor and calibration status directly.
- Analysis and expert views additionally show dead time, correction model, reliable CPM limit, uncertainty values and calibration reference.

GMC-500+ dual-tube profiles

- M4011 and SI-3BG can use separate conversion factors, dead times and reliable count-rate ranges.
- Separate mode uses the two tube channels. Curve mode applies a piecewise dual-tube calibration curve.
- When the SI-3BG calibration is missing, the app does not calculate a misleading dose value with the M4011 factor.

Note: Bundled default values are working values and do not replace a traceable calibration certificate.

Addendum for version 8.3.5

This edition supplements the existing user manual with the changes introduced in version 8.3.5.

Detector presets and unambiguous tube configuration

- GMC-320 Plus V4: one physical M4011 tube. The app automatically loads 154 CPM/(µSv/h), 120 µs dead time, the non-paralyzable model, a 50,000 CPM working limit and 20% factor uncertainty.
- GMC-500+: separate profiles for the primary M4011 tube and the second SI-3BG tube. No dose factor is invented for the SI-3BG without verified device calibration.
- Duplicated low_dose_* fields were removed from the Home Assistant configuration. The normal calibration fields now describe the only or primary tube.
- The dashboard shows one detector profile for single-tube devices and two separate profiles for dual-tube devices.
- Existing version 8.3.4 configurations are migrated automatically and remain backward compatible.

> Note: The predefined values are documented application working values and do not replace a traceable calibration certificate.

Version 8.4.1 addendum

Detector, calibration and scientific traceability

- Automatic profile detection for GMC-300, GMC-320, GMC-500+ and GMC-600+.
- New expert card with tube, calibration status, dead time and active detector.
- Live measurement quality, detector load, estimated losses and correction factor.
- GMC-500+: separate M4011 and SI-3BG display; no invented SI-3BG dose factor.
- Calibration profiles, assistant and versioned calibration history.
- Raw, dead-time-corrected and comparison modes for charts and reports.
- Optional scientific PDF report with an additional metrology page.

Note: predefined working values do not replace a traceable calibration certificate.

Addendum for version 8.4.2

Interface and device configuration

Version 8.4.2 removes the duplicate measurement-quality and device-comparison tiles. The detector profile now contains only supplementary information; raw values, dead-time correction and the quality index remain traceable in Expert view.

The GMC-320 is configured only as a single-tube device. The GMC-500+ has exactly one complete dual-tube entry for M4011 and SI-3BG. Existing split 8.4.0/8.4.1 entries are merged losslessly at startup.

Single tube GMC-300 / GMC-320	Dual tube GMC-500+ (M4011 + SI-3BG)